Typography & Web Fonts

1- Google Fonts, 1,500+ free professional fonts
2- Font Properties - size, weight, spacing, style
3- Line Height & Spacing - making text readable
4- Type Hierarchy - creating visual structure
5- Font Pairing - combining fonts like a pro

Ready to make your text look professional?



SCROLL

Why Typography Matters

Typography = 90% of Web Design
Why? Because most of what users see and read is TEXT!

Good Typography:

- Makes content readable - Users can read your message
- Creates hierarchy - Shows what's important vs secondary
- Establishes mood - Formal? Playful? Professional? Creative?
- Looks professional - Instantly elevates your design
- Keeps users engaged - Comfortable reading = longer visits
- Builds trust - Clean typography = credible content

Bad Typography:

- Drives users away immediately
- Makes content hard to read
- Looks amateurish
- Confuses hierarchy (what to read first?)
- Hurts your message

The Truth: You can have amazing content,

but if the typography is bad, no one will read it.

Font Families Overview

1. Serif - Traditional, readable, formal

- Examples: Times New Roman, Georgia, Merriweather
- Has small lines (serifs) at the ends of letters

2. Sans-Serif - Modern, clean, versatile

- Examples: Arial, Helvetica, Open Sans, Roboto
- No serifs - smooth, clean lines

3. Monospace - Fixed-width, technical

- Examples: Courier, Consolas, Source Code Pro
- Every character takes the same width

4. Cursive - Handwriting-style, decorative

- Examples: Pacifico, Dancing Script
- Use VERY sparingly!

5. Fantasy - Highly decorative, themed

- Examples: Impact, Papyrus
- Use for special effects only

Serif Fonts

What are serifs?
Small decorative lines (feet) at the ends of letter strokes

Popular Serif Fonts:

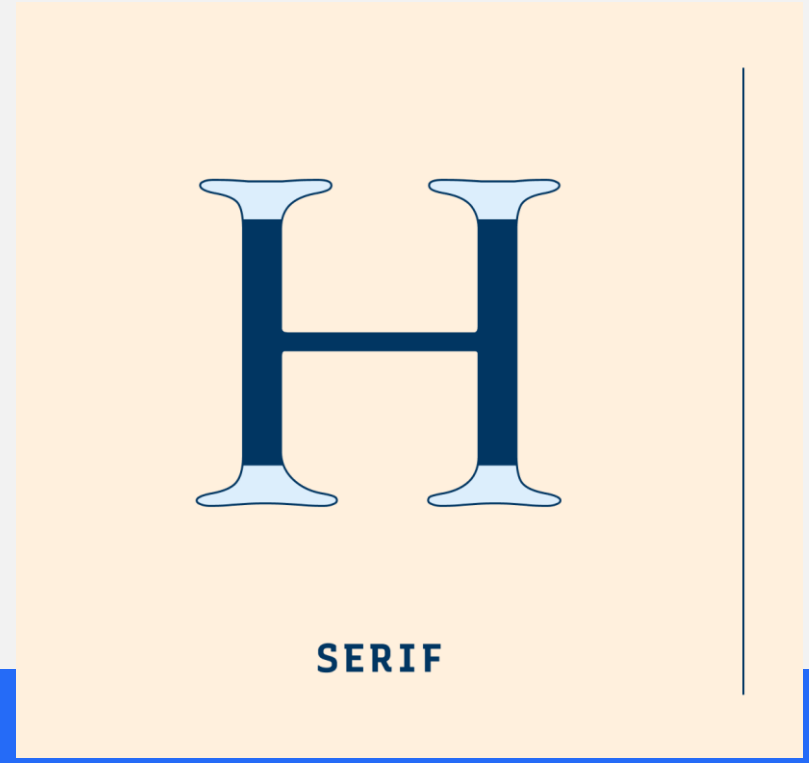
- Times New Roman - Classic, traditional
- Georgia - Web-friendly, highly readable
- Merriweather - Modern serif, designed for screens
- Playfair Display - Elegant, editorial style
- Crimson Text - Book-like, sophisticated

Best Used For:

- Long-form reading (articles, blogs)
- Traditional/formal feel (law firms, universities)
- Print-like appearance (news sites, magazines)
- Body text in editorial designs
- Establishing credibility and trust

Personality/Mood:

- Classic, traditional, trustworthy
- Formal, established, authoritative
- Literary, editorial, sophisticated



Famous Sites Using Serif:
Medium (for article text); The New York; Times Time Magazine

Sans-Serif Fonts

Popular Serif Fonts:

- Arial - Universal standard (safe choice)
- Helvetica - Classic, professional
- Open Sans - Friendly, highly readable
- Roboto - Google's modern standard
- Montserrat - Geometric, stylish
- Inter - Designed for UI/screens

Best Used For:

- Modern websites (most common choice!)
- UI elements (buttons, navigation, forms)
- Headings and titles
- Digital screens (very readable)
- Clean, minimal designs

Personality/Mood:

- Modern, clean, contemporary
- Approachable, friendly, accessible
- Professional without being stuffy
- Minimalist, straightforward

What is Sans-Serif?
"Sans" = Without (French) No decorative lines - clean, simple strokes only



Famous Sites Using Sans-Serif:
Google (all products); Apple; Facebook;
Most modern websites!

Monospace Fonts

Popular Serif Fonts:

- Courier New - Classic typewriter font
- Consolas - Windows standard for code
- Monaco - macOS standard for code
- Source Code Pro - Adobe's coding font
- Fira Code - Modern, includes coding ligatures

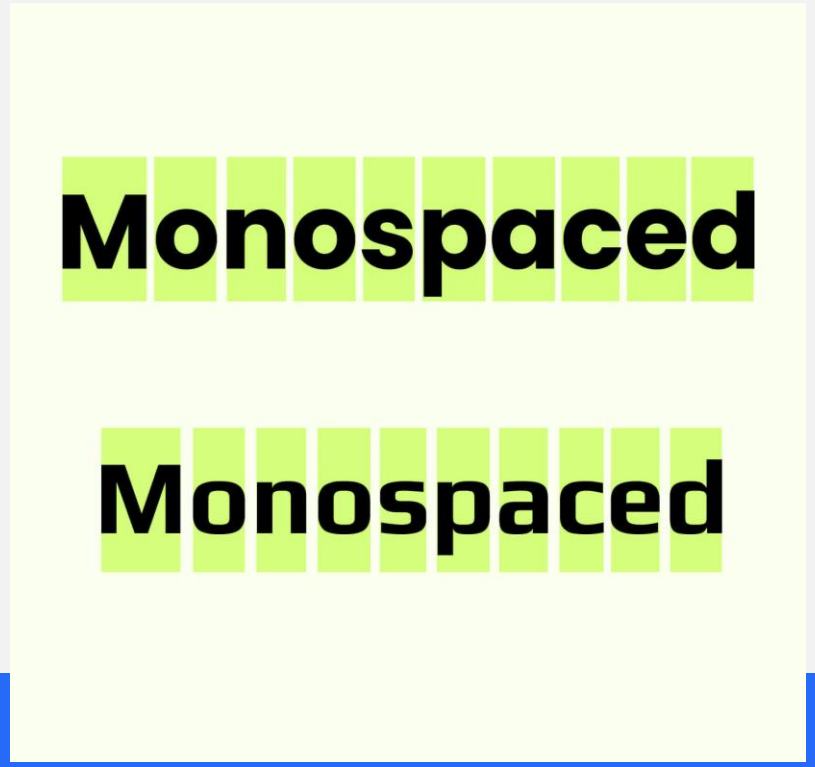
Best Used For:

- Code snippets and examples
- Technical documentation
- Data tables and aligned text
- Terminal/command line displays
- Displaying file paths or URLs

When NOT to Use:

- Body text (hard to read in paragraphs)
- Headings (looks too technical)
- Regular content (unless intentionally retro)

What is Monospace? Every character has the SAME width (like a typewriter)



These are for code display. Each character takes the same space

Font Family CSS

```css
body {
    font-family: Arial, Helvetica, sans-serif;
}
```

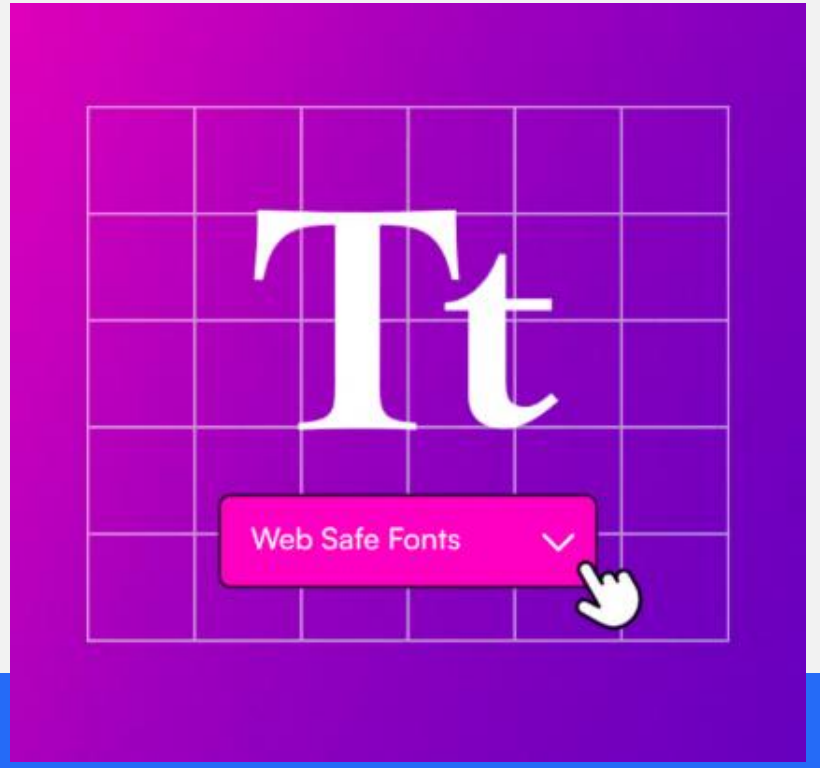
This is a "Font Stack" - How It Works:

1. Try Arial first (not installed?)
2. Try Helvetica (not installed?)
3. Use any sans-serif font

The 5 Generic Font Families:

- serif - Any serif font
- sans-serif - Any
- sans-serif font
- monospace - Any monospace font
- cursive - Any cursive font
- fantasy - Any decorative font

Why? Not everyone has the same fonts installed on their computer!



Golden Rule: ALWAYS end with a generic family!

Font Size

Font Size Units in CSS. Three Main Ways to Size Text:

```css
p {
    font-size: 16px;  /* Always exactly 16 pixels */
}

h1 {
    font-size: 48px;  /* Always exactly 48 pixels */
}
```

1. Pixels (px) - Fixed Size

- Simple and precise
- Doesn't scale with user preferences
- Old-school approach

```css
body {
    font-size: 16px;
}

p {
    font-size: 1.5em;  /* 1.5 × 16px = 24px */
}

span {
    font-size: 0.875em;  /* 0.875 × parent size */
}
```

2. Em - Relative to Parent

- Scales based on parent element
- Can compound (gets confusing!)

Font Size - Cont.

```css
html {
    font-size: 16px;  /* Root size */
}

body {
    font-size: 1rem;  /* 1 × 16px = 16px */
}

h1 {
    font-size: 3rem;  /* 3 × 16px = 48px */
}

p {
    font-size: 1rem;  /* 1 × 16px = 16px */
}

small {
    font-size: 0.875rem;  /* 0.875 × 16px = 14px */
}
```

Recommendation: Use rem for font sizes, it's the modern standard!

3. Rem - Relative to Root

- Scales based on root element (html)
- Predictable and consistent
- Modern best practice!

PX vs EM vs REM

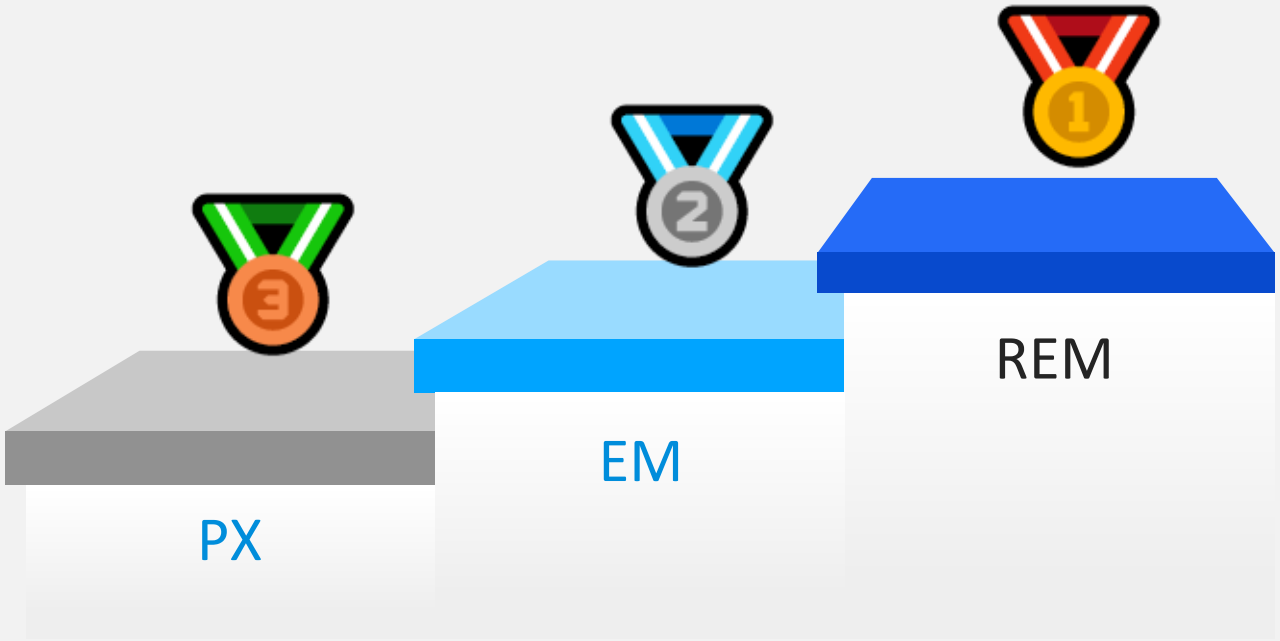
REM (Root EM)

- Scales relative to root (html) only
- Predictable and consistent everywhere
- No compounding issues!
- Respects user browser settings (accessible!)
- Modern standard

01 Use **REM** for font sizes (modern, accessible, predictable)

02 Use **EM** for spacing that should scale with text (padding, margin)

03 Use **PX** sparingly (borders, shadows, small details)



Font Weight

```css
p {
    font-weight: normal;  /* Regular text */
    font-weight: bold;    /* Bold text */
}

h1 {
    font-weight: 100;  /* Thin */
    font-weight: 300;  /* Light */
    font-weight: 400;  /* Normal/Regular */
    font-weight: 500;  /* Medium */
    font-weight: 600;  /* Semi-bold */
    font-weight: 700;  /* Bold */
    font-weight: 900;  /* Extra/Black */
}
```

Pro Tip: Use numeric values (400, 700) for more control and precision!

Important Notes:

- Not all fonts have all weights!
- Some fonts only have 400 (normal) and 700 (bold)
- Google Fonts shows which weights are available
- If you use 600 but font doesn't have it, browser picks closest available

Font Weight - Cont.

Common Combinations ->

```
/* Body text */
body {
    font-weight: 400;  /* Normal, easy to read */
}

/* Headings */
h1, h2, h3 {
    font-weight: 700;  /* Bold, stands out */
}

/* Subtle emphasis */
.subtitle {
    font-weight: 300;  /* Light, elegant */
}

/* Strong emphasis */
strong {
    font-weight: 700;  /* Bold */
}

/* Button text */
button {
    font-weight: 600;  /* Semi-bold, clear */
}
```

Font Style

```css
/* Font style */
em {
    font-style: italic; /* Slanted, true italic */
}

.quote {
    font-style: italic;
}

.normal {
    font-style: normal; /* Remove italic */
}

.oblique {
    font-style: oblique;/* Slanted (fake italic) */
}
```

Font style is mainly for italics - useful for emphasis, quotes, or styling the tag

What's the difference?

- italic = Uses font's designed italic version (better quality)
- oblique = Browser slants regular letters (artificial)
- Always use italic when available!

Text Transform

```css
/* Uppercase - ALL CAPS */
h1 {
    text-transform: uppercase;  /* MAKES TEXT LIKE THIS */
}

/* Lowercase - all lowercase */
.lowercase {
    text-transform: lowercase;  /* makes text like this */
}

/* Capitalize - First Letter Of Each Word */
.title {
    text-transform: capitalize; /* Makes Text Like This */
}

/* Remove transform */
.normal {
    text-transform: none;       /* Text as written in HTML */
}
```

When to Use What:

text-transform: uppercase

- Navigation menus
- Buttons and CTAs
- Section labels
- Emphasis without bold
- Add letter-spacing for readability!

text-transform: capitalize

- Titles and headings
- Names in databases
- Consistent title formatting

text-transform: capitalize

- Email addresses
- Usernames
- Tags and labels

Google Fonts Revolution

Why It's Revolutionary:

- 100% FREE - No licenses, no fees, forever
- 1,500+ fonts - Massive selection
- Professional quality - Designed by experts
- Easy to use - 3 simple steps to add to your site
- Fast loading - Hosted on Google's CDN (Content Delivery Network)
- Works everywhere - All browsers, all devices
- Used by millions - Trusted by professional designers

Before Google Fonts (2010):

- Limited to "web-safe fonts" (Arial, Times, Georgia)
- Or pay $$ for font licenses
- Or complicated self-hosting

After Google Fonts:

- Thousands of free, beautiful fonts
- Add in seconds
- No technical complexity



Google Fonts

How to Use Google Fonts

1- Browse & Select

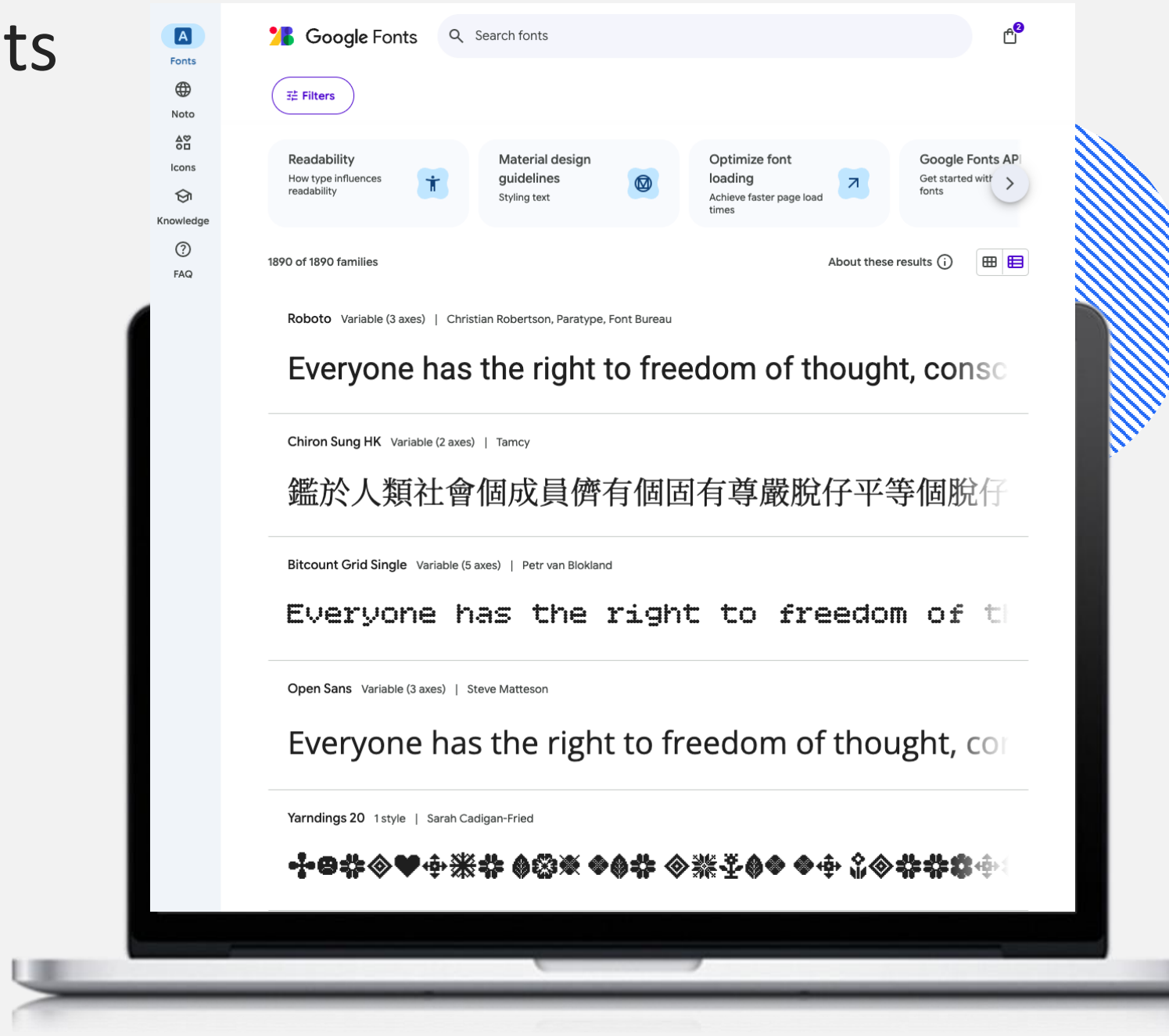
- Go to fonts.google.com
- Browse the 1,500+ fonts
- Click fonts you like
- Select the weights you need (400, 700, etc.)

2- Get the Code

- Google generates code for you
- Copy the <link> tags
- Copy the CSS font-family rules

3- Add to Your Site

- Paste <link> tags in your HTML <head>
- Use font-family in your CSS
- Save, refresh, done!



Getting the Code

Part 1: HTML <link> Tags (Copy These!)

```
<link rel="preconnect" href="https://fonts.googleapis.com">
<link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
<link href="https://fonts.googleapis.com/css2?family=Montserrat:ital,wght
    @0,100..900;1,100..900&family=Open+Sans:ital,wght@0,300..800;1,300..800&
    display=swap" rel="stylesheet">
```

What these do: Connect your site to Google's font servers
Where they go: In your HTML <head> section

Part 2: CSS Font-Family Rules (Copy These!)

```
font-family: 'Montserrat', sans-serif;
font-family: 'Open Sans', sans-serif;
```

What these do: Tell CSS which font to use
Where they go: In your CSS file

Applying Google Fonts to Your Site – Part 1

Find the <head> section and paste the link tags:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>My Website</title>

    <!-- PASTE GOOGLE FONTS LINKS HERE -->
    <link rel="preconnect" href="https://fonts.googleapis.com">
    <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
    <link href="https://fonts.googleapis.com/css2?
        family=Montserrat:wght@400;600;700&family=Open+Sans:ital,
        wght@0,400;0,700;1,400&display=swap" rel="stylesheet">

    <link rel="stylesheet" href="style.css">
</head>
```

Applying Google Fonts to Your Site – Part 2

Apply fonts to your elements:

```css
/* Body text - Open Sans */
body {
    font-family: 'Open Sans', sans-serif;
    font-size: 16px;
    line-height: 1.6;
    color: #333;
}

/* All headings - Montserrat */
h1, h2, h3, h4, h5, h6 {
    font-family: 'Montserrat', sans-serif;
    font-weight: 700;
}

/* Optional: Different weight for subheadings */
h2 {
    font-weight: 600;
}
```

Troubleshooting:

- Are ALL 3 link tags in HTML <head>?
- Are link tags BEFORE style.css link?
- Did you save BOTH HTML and CSS files?
- Did you refresh the browser? (Ctrl+F5 or Cmd+Shift+R)
- Are font names spelled exactly right with quotes?
  - `font-family: 'Open Sans', sans-serif;`

Font Pairing Tips

The Golden Rule: Contrast + Harmony

Fonts should be different enough to create visual interest, but similar enough to work together.

Classic Pairing Strategies:

1. Display + Simple Heading:

- Bold, decorative, eye-catching font
- Body: Clean, simple, readable font

2. Serif + Sans-Serif:

- Heading: Serif (traditional, authoritative)
- Body: Sans-serif (modern, clean)

3. Geometric + Humanist:

- Heading: Geometric sans (modern, structured)
- Body: Humanist sans (friendly, readable)

Proven Font Combinations:

Classic & Elegant:

- Playfair Display + Lato
- Cormorant Garamond + Proza Libre
- Crimson Text + Work Sans

Modern & Bold:

- **Montserrat + Open Sans (Today's choice!)**
- Oswald + Source Sans Pro
- Poppins + Inter

Traditional & Trustworthy:

- Roboto Slab + Roboto
- Merriweather + Open Sans
- Libre Baskerville + Source Sans Pro

Playful & Creative:

- Righteous + Lato
- Fredoka One + Nunito

Font Pairing Tips – Cont.

Font Pairing Rules:

DO:

- Limit to 2-3 font families MAX
- Create clear contrast (different but compatible)
- Use one font for headings, another for body
- Match the mood (formal? playful? professional?)
- Test readability at different sizes

DON'T:

- Use too many fonts (looks chaotic!)
- Pair two decorative fonts (too busy)
- Pair fonts that are TOO similar (no contrast)
- Use fonts that clash in style
- Sacrifice readability for style

fontpair.co fontjoy.com type-scale.com

```css
/* Heading font: Bold & distinctive */
h1, h2, h3 {
    font-family: 'Montserrat', sans-serif;
    font-weight: 700;
}

/* Body font: Clean & readable */
body {
    font-family: 'Open Sans', sans-serif;
    font-weight: 400;
}
```

Pro Tips: Steal from the pros - Look at sites you admire, inspect their fonts (DevTools!)

Line Height

What is Line Height? The vertical space between lines of text
(Called "leading" in traditional typography - pronounced "ledding")

CSS Syntax:

```css
p {
    line-height: 1.6;  /* 1.6 times the font size */
}

/* Can also use units */
p {
    line-height: 24px;  /* Fixed */
    line-height: 1.5em; /* Relative */
}
```

How It Works:
font-size: 16px line-height: 1.6 → 1.6 × 16px = 25.6px between lines

Recommended Line Height Values:
Body Text (Paragraphs):

- **1.4 - 1.8** (comfortable reading)
- Sweet spot: **1.6** (most common)

Headings:

- **1.1 - 1.3** (tight, impactful)
- Larger headings can go tighter

Small Text:

- **1.4 - 1.5** (needs more space)

Pro Tips:

- Use unitless values (1.6, not 1.6em) - cleaner inheritance
- Longer lines need more line height
- Narrow columns can use tighter line height
- Line height affects overall page rhythm

Letter Spacing

What is Letter Spacing? The horizontal space between individual letters/characters (Called "tracking" in traditional typography)

CSS Syntax:

```css
h1 {
    letter-spacing: 2px;    /* Add 2px between letters */
}

.tight {
    letter-spacing: -1px;  /* Negative = tighter */
}

.spaced {
    letter-spacing: 0.1em; /* Relative to font size */
}

/* Default */
.normal {
    letter-spacing: normal;/* or 0 */
}
```

DO:

- Add 1-2px to UPPERCASE text
- Use subtle adjustments (-1px to 2px)
- Test readability at different sizes
- Use on headings, buttons, labels

DON'T:

- Add to body paragraphs
- Use extreme values (5px+)
- Forget to adjust when using uppercase

Pro Tips:

- Golden Rule: UPPERCASE = add letter-spacing (always!)
- Subtlety wins - most adjustments should be -1px to 2px
- Negative spacing can create tight, modern headlines
- Use sparingly - too much destroys readability

Text Alignment

CSS Syntax:

```css
/* 1. Left Align ← DEFAULT (Best!) */
text-align: left;

/* 2. Center */
text-align: center;

/* 3. Right Align */
text-align: right;

/* 4. Justify */
text-align: justify;
```

The Golden Rule:
Left-align body text for maximum readability!

Left Align:
- Body text, paragraphs, lists
- Most readable option
- Default for a reason!

Center:
- Headings, hero sections
- Short phrases (1-3 lines only)
- NOT for body paragraphs!

Right Align:
- Dates, prices, metadata
- Rarely used in English

Justify:
- Both edges aligned
- Can create awkward spacing
- Use carefully!

Creating Type Hierarchy

Three Tools to Create Hierarchy:

```css
/* 1. Size (Most Powerful) */
h1 { font-size: 3rem; }    /* 48px - Biggest */
h2 { font-size: 2.5rem; }  /* 40px */
h3 { font-size: 2rem; }    /* 32px */
p  { font-size: 1rem; }    /* 16px - Smallest */

/* 2. Weight (Emphasis) */
h1 { font-weight: 700; }   /* Bold - Strong */
h2 { font-weight: 600; }   /* Semi-bold */
p  { font-weight: 400; }   /* Normal */

/* 3. Margin (Spacing) */
h1 { margin-bottom: 1rem; }
h2 { margin-top: 2rem; margin-bottom: 1rem; }
p  { margin-bottom: 1.5rem; }
```

Why Hierarchy Matters:

- Guides the eye (where to look first)
- Shows importance (what matters most)
- Improves readability (easier to scan)
- Looks professional (organized, intentional)

Good (Clear Hierarchy):

- Obvious size differences
- Clear importance levels
- Professional appearance
- Easy to scan

The Rule:
Each heading level should be NOTICEABLY different!

Type Hierarchy Example

The Result: Clear visual hierarchy where each level is OBVIOUSLY different!

```css
/* Reset and Base Styles */
* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}

body {
    font-family: 'Open Sans', sans-serif;
    font-size: 16px;
    line-height: 1.6;
    color: #333;
    background-color: #fff;
}

/* Heading Hierarchy */
h1, h2, h3, h4, h5, h6 {
    font-family: 'Montserrat', sans-serif;
    line-height: 1.2;
    color: #2c3e50;
}

h1 {
    font-size: 3rem;      /* 48px - Hero/Page Title */
    font-weight: 700;
    margin-bottom: 1rem;
}

h2 {
    font-size: 2.5rem;    /* 40px - Major Sections */
    font-weight: 600;
    margin-top: 2rem;
    margin-bottom: 1rem;
}

h3 {
    font-size: 2rem;      /* 32px - Subsections */
    font-weight: 600;
    margin-top: 1.5rem;
    margin-bottom: 0.75rem;
}

h4 {
    font-size: 1.5rem;    /* 24px - Minor Headings */
    font-weight: 600;
}

/* Body Text */
p {
    margin-bottom: 1.5rem;
}

/* Emphasis */
strong {
    font-weight: 700;
}

em {
    font-style: italic;
}
```

Typography Best Practices

DO:

Font Families:

- Limit to **2 font families max** (one for headings, one for body)
- Choose readable fonts

Hierarchy:

- Create **clear size differences** between heading levels
- Use consistent spacing (multiples: 1rem, 1.5rem, 2rem)

Readability:

- Keep line length **45-75 characters** per line
- Use **line-height: 1.6** for body text
- Ensure **good contrast** (dark text on light background)

Alignment:

- **Left-align body text** for readability
- Center only short headings (1-3 lines)

Size:

- Body text minimum **16px** (never smaller!)

DON'T:

Font Mistakes:

- Use more than 3 font families (looks chaotic!)
- Use decorative fonts for body text

Alignment Mistakes:

- Center long paragraphs (hard to read)
- Justify without proper line length

Readability Mistakes:

- Low contrast colors (light gray on white)
- Body text smaller than 16px (too hard to read)
- Lines longer than 90 characters (eye gets lost)

Spacing Mistakes:

- Inconsistent spacing (random margins)
- No spacing between sections (cramped)

Line Length

Optimal Line Length: The Magic Number: 45-75 Characters Per Line

Using Characters (ch unit) - BEST METHOD!

```
main {
    max-width: 65ch; /* 65 characters wide */
    margin: 0 auto;  /* Center it */
}

article {
    max-width: 70ch; /* ~70 characters */
}
```

What is 'ch'?

- 1ch = width of the "0" character in the current font
- Perfect for controlling text width!

Pro Tips:

- Use 'ch' unit - it scales with font size automatically!
- 65ch is the sweet spot for most body text
- Always center with margin: 0 auto
- Add padding for mobile - prevents text touching edges

Quick Rule: If you turn your head to read, the line is too long!


Google Fonts and how to use them?
Selectors or any CSS syntax?
HAVE ANY
QUESTIONS?
Typography properties (line-height, letter-spacing)?
The Typography Challenge assignment?
The Box Model (padding, margin, border)?
Anything from this week?